CM5 socket

File: cm5_socket_mcu.kicad_sch

STM32H7 MCU

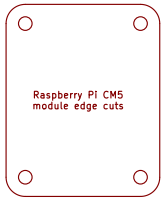
File: STM32H7_MCU.kicad_sch

Ethernet PHY, PSRAM, SDcard

File: eth_psram_emmc.kicad_sch

DVID converter, NOR Flash

File: Ethernet_flash.kicad_sch



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32-bit STM32 MCU module

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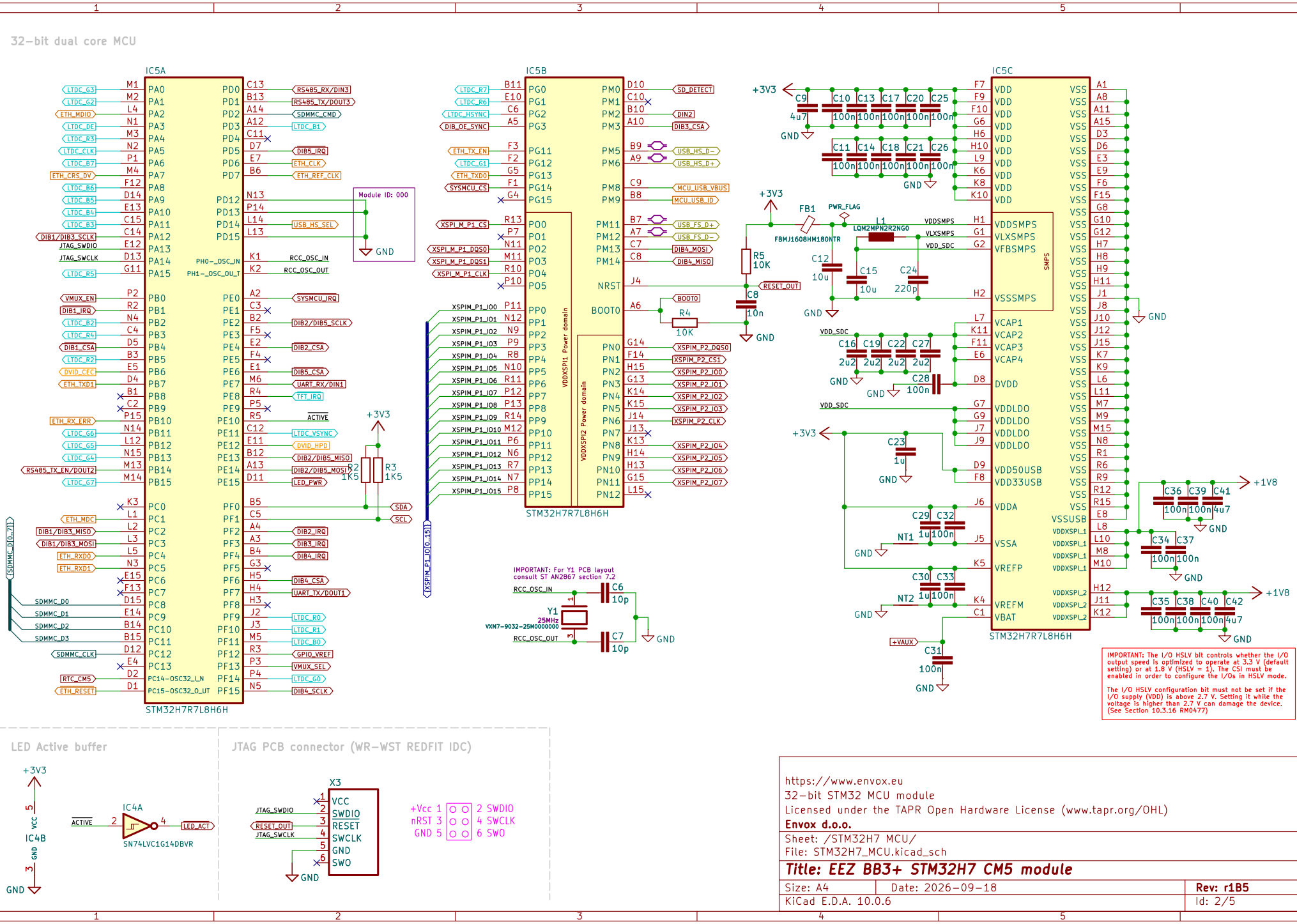
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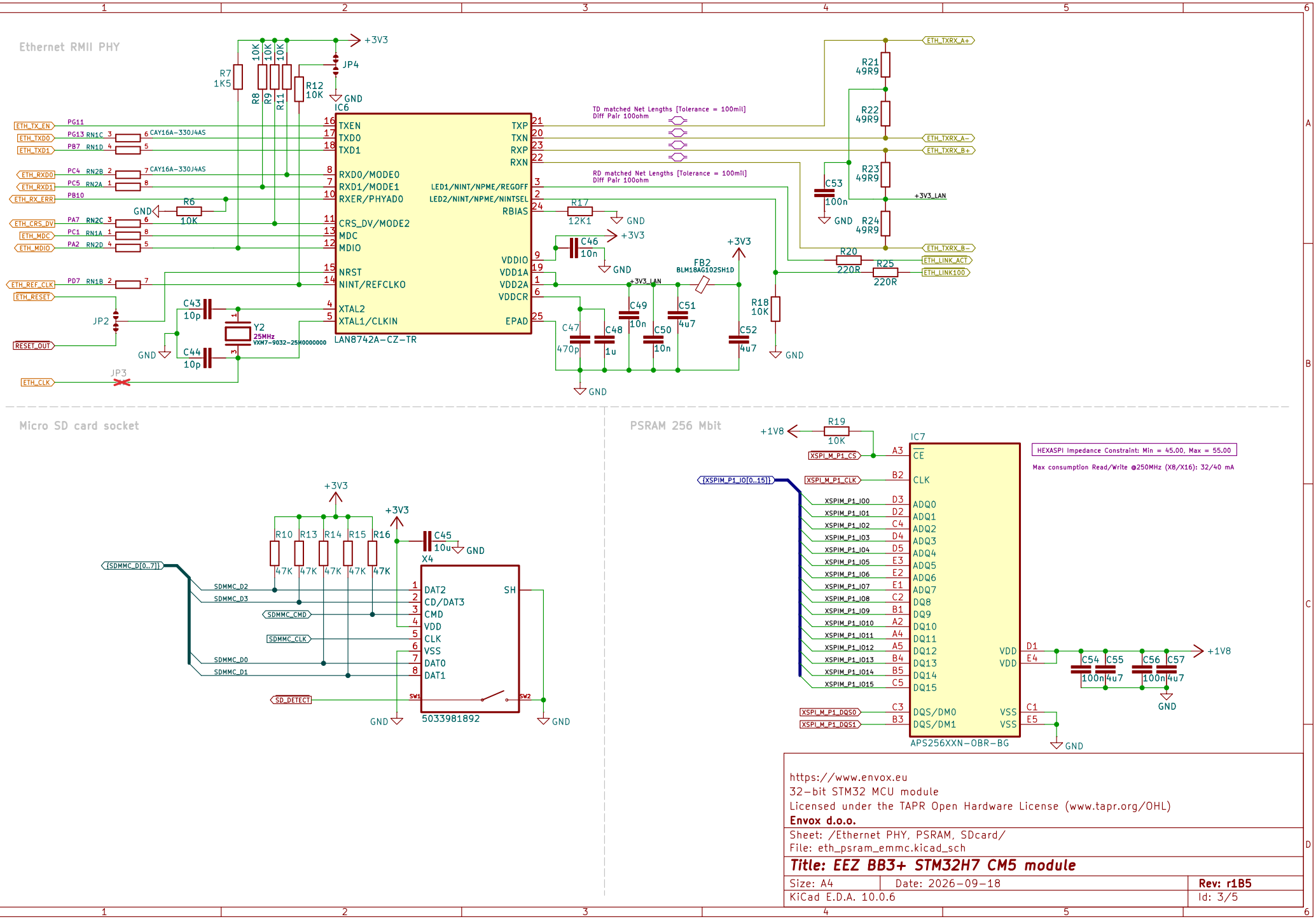
Sheet: /

File: BB3plus CM5 STM32H7 module.kicad_sch

Title: EEZ BB3+ STM32H7 CM5 module

Size: A4	Date: 2026-09-18	**Rev: r1B5**
KiCad E.D.A. 10.0.6		Id: 1/5


32-bit dual core MCU
IC5A
STM32H7R7L8H6
IC5B
STM32H7R7L8H6
IC5C
STM32H7R7L8H6
Module ID: 000
IMPORTANT: For Y1 PCB layout consult ST AN2867 section 7.2
IMPORTANT: The I/O HSLV bit controls whether the I/O output speed is optimized to operate at 3.3 V (default setting) or at 1.8 V (HSLV = 1). The CSI must be enabled in order to configure the I/Os in HSLV mode.
The I/O HSLV configuration bit must not be set if the I/O supply (VDD) is above 2.7 V. Setting it while the voltage is higher than 2.7 V can damage the device. (See Section 10.3.16 RM0477)
LED Active buffer
JTAG PCB connector (WR-WST REDFIT IDC)
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Sheet: /STM32H7 MCU/
File: STM32H7_MCU.kicad_sch
Title: EEZ BB3+ STM32H7 CM5 module
Size: A4
Date: 2026-09-18
Rev: r1B5
KiCad E.D.A. 10.0.6
Id: 2/5


Ethernet RMII PHY
IC6
LAN8742A-CZ-TR
TXEN
TXD0
TXD1
RXD0/MODE0
RXD1/MODE1
RXER/PHYAD0
CRS_DV/MODE2
MDC
MDIO
NRST
NINT/REFCLKO
XTAL2
XTAL1/CLKIN
TXP
TXN
RXP
RXN
LED1/NINT/NPME/REGOFF
LED2/NINT/NPME/NINTSEL
RBIAS
VDDIO
VDD1A
VDD2A
VDDCR
EPAD
TD matched Net Lengths [Tolerance = 100mil]
Diff Pair 100ohm
RD matched Net Lengths [Tolerance = 100mil]
Diff Pair 100ohm
ETH_TXRX_A+
ETH_TXRX_A-
ETH_TXRX_B+
ETH_TXRX_B-
ETH_LINK_ACT
ETH_LINK100
ETH_TX_EN
ETH_TXD0
ETH_TXD1
ETH_RXD0
ETH_RXD1
ETH_RX_ERR
ETH_CRS_DV
ETH_MDC
ETH_MDIO
ETH_REF_CLK
ETH_RESET
RESET_OUT
ETH_CLK
CAY16A-330J4AS
FB2
BLM18AG102SH1D
Y2
25MHz
VXM7-9032-25M0000000
+3V3
+3V3_LAN
GND
JP2
JP3
JP4
Micro SD card socket
X4
5033981892
DAT2
CD/DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
SH
SW1
SW2
SDMMC_D2
SDMMC_D3
SDMMC_CMD
SDMMC_CLK
SDMMC_D0
SDMMC_D1
SD_DETECT
[SDMMC_D[0..7]]
PSRAM 256 Mbit
IC7
APS256XXN-OBR-BG
HEXASPI Impedance Constraint: Min = 45.00, Max = 55.00
Max consumption Read/Write @250MHz (X8/X16): 32/40 mA
[XSPIM_P1_IO[0..15]]
XSPI_M_P1_CS
XSPI_M_P1_CLK
XSPI_M_P1_DQS0
XSPI_M_P1_DQS1
+1V8
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Sheet: /Ethernet PHY, PSRAM, SDcard/
File: eth_psram_emmc.kicad_sch
Title: EEZ BB3+ STM32H7 CM5 module
Size: A4
Date: 2026-09-18
Rev: r1B5
KiCad E.D.A. 10.0.6
Id: 3/5

RGB-to-DVID converter

Video MUX 1:2

NOR flash memory 256 Mbit (32 MB x 8, 200 MHz)

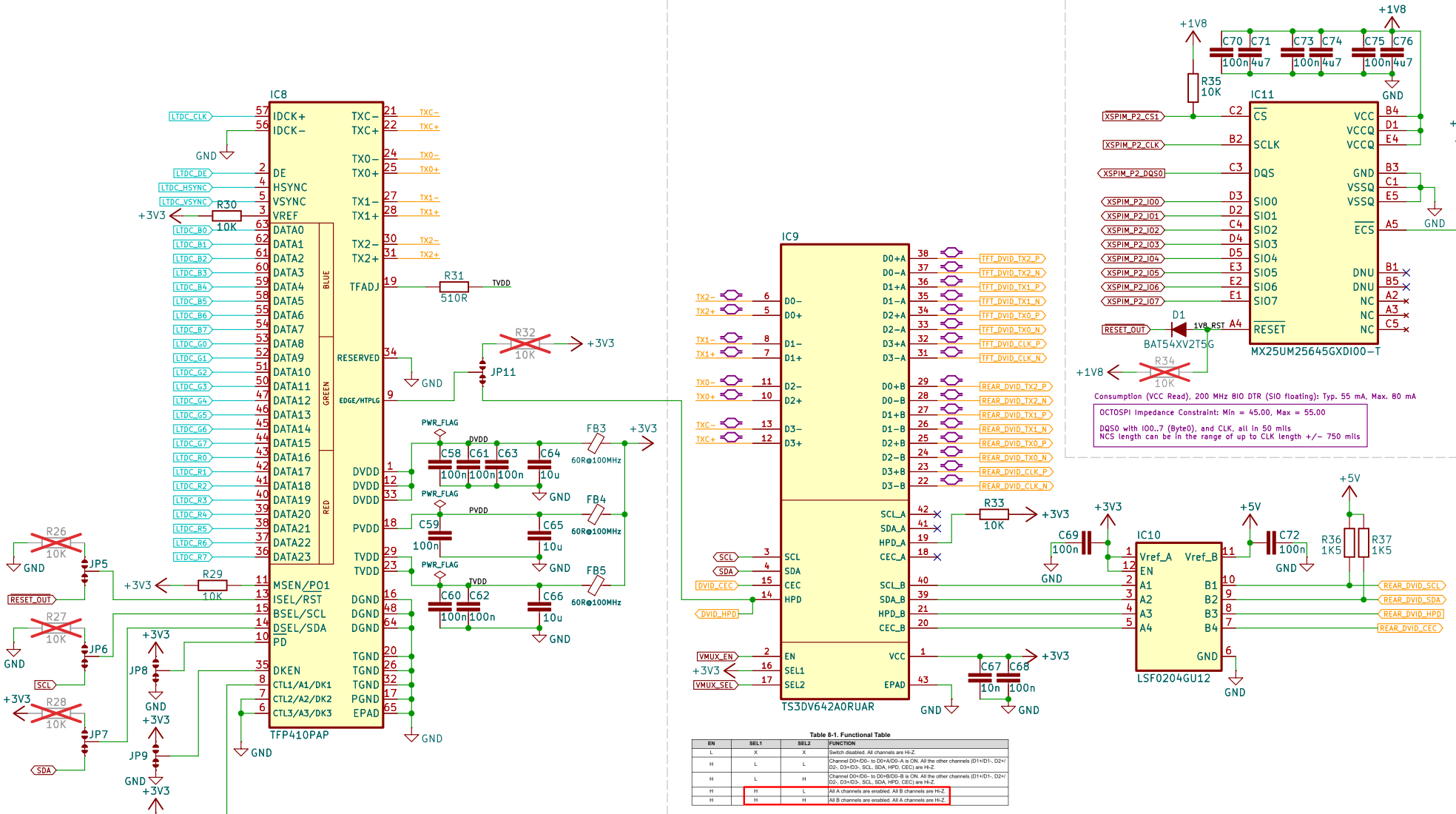


Table 8-1. Functional Table

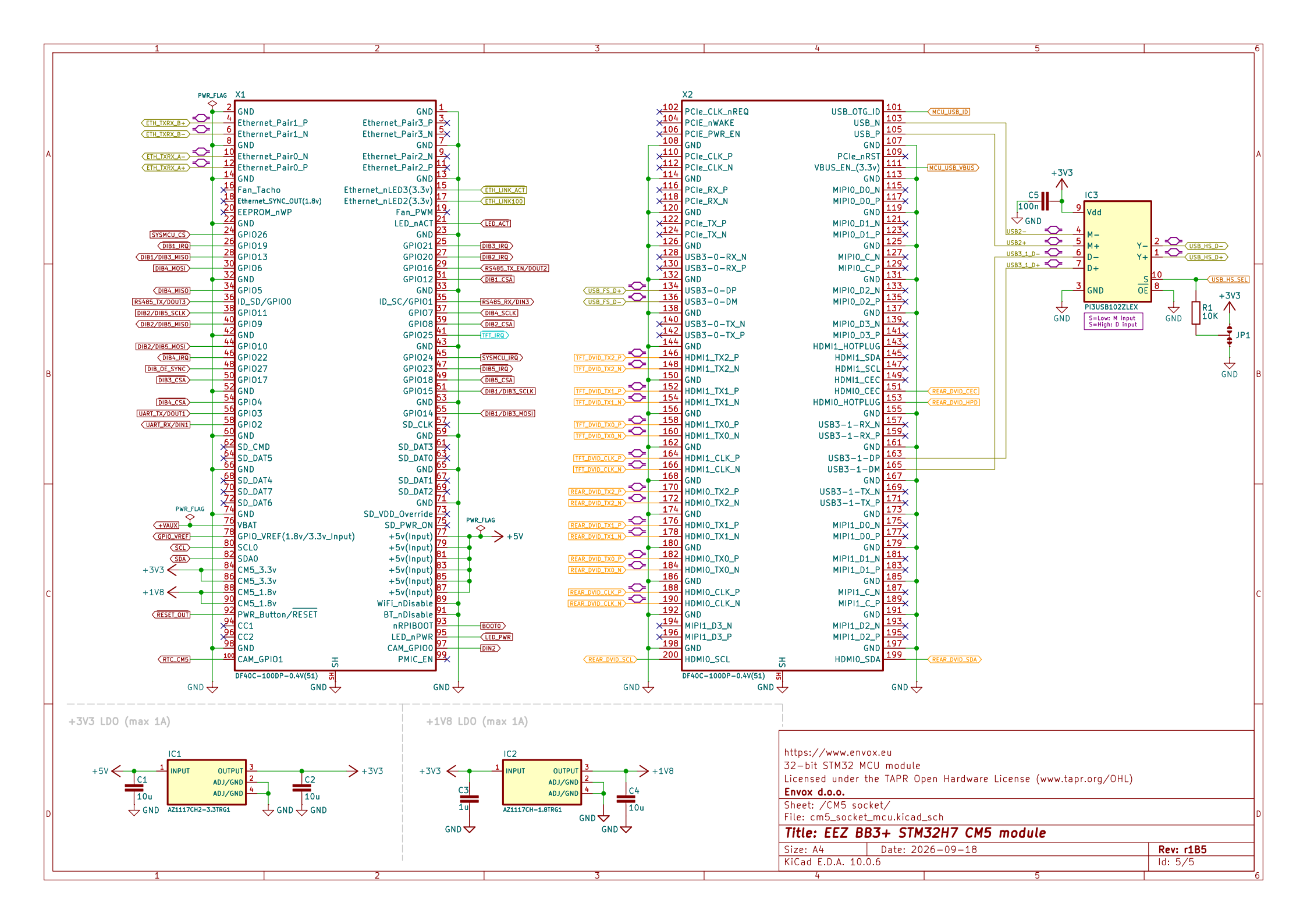
EN	SEL1	SEL2	FUNCTION
L	X	X	Switch disabled. All channels are Hi-Z.
H	L	L	Channel D0+/D0- to D0+A/D0-A is ON. All the other channels (D1+/D1-, D2+/D2-, D3+/D3-, SCL, SDA, HPD, CEC) are Hi-Z.
H	L	H	Channel D0+/D0- to D0+B/D0-B is ON. All the other channels (D1+/D1-, D2+/D2-, D3+/D3-, SCL, SDA, HPD, CEC) are Hi-Z.
H	H	L	All A channels are enabled. All B channels are Hi-Z.
H	H	H	All B channels are enabled. All A channels are Hi-Z.

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Sheet: /DVID converter, NOR Flash/
File: Ethernet_flash.kicad_sch

Title: EEZ BB3+ STM32H7 CM5 module

Size: A4	Date: 2026-09-18	**Rev: r1B5**
KiCad E.D.A. 10.0.6		Id: 4/5


X1
X2
DF40C-100DP-0.4V(51)
IC1
AZ1117CH2-3.3TRG1
IC2
AZ1117CH-1.8TRG1
IC3
PI3USB102ZLEX
S=Low: M input
S=High: D input
+3V3 LDO (max 1A)
+1V8 LDO (max 1A)
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Envox d.o.o.
Sheet: /CM5 socket/
File: cm5_socket_mcu.kicad_sch
Title: EEZ BB3+ STM32H7 CM5 module
Size: A4
Date: 2026-09-18
Rev: r1B5
KiCad E.D.A. 10.0.6
Id: 5/5